

Abnormal Heartbeat Audio Detection Comparison between a Convolutional Neural Network on Mel-Spectrograms and Gradient Boosting on Audio Features

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ABSTRACT

Automated detection of heartbeat abnormalities from audio recordings can enhance cardiological screening and reduce clinician workload. This study compares two machine learning approaches: a convolutional neural network (CNN) trained on Mel-spectrograms and a gradient boosting classifier (XGBoost) trained on engineered audio features. Models were evaluated on a held-out evaluation set and two independent test datasets: mobile recordings and clinical phonocardiograms representing diverse acquisition conditions. Although gradient boosting would have served as a baseline model for comparison, it outperformed the convolutional neural network in almost all tests. On the evaluation set, XGBoost achieved the highest performance (high accuracy, balanced precision and recall), while CNN overly favored abnormal beats and struggled with normal beats. Ensemble predictions mirrored the gradient boosting model, indicating partial complementarity but limited added value from CNN. On mobile app recordings, CNN maintained moderate accuracy with high recall on abnormalities, whereas XGBoost performed poorly. The ensemble slightly improved class balance. On clinical recordings, both models achieved similarly low accuracy, with minimal ensemble gain, partly suggesting robustness on high-quality clinical data. These results highlight that feature-based models can outperform convolutional neural networks on spectrograms for heartbeat detection, and ensemble models can stabilize and improve predictions for noisy settings.

1. INTRODUCTION

Accurate detection of heartbeat abnormalities using machine learning techniques has the potential to substantially reduce the workload of physicians and increase the efficiency of cardiological examinations by highlighting cases that require further clinical attention. With the widespread adoption of digital stethoscopes, the initial stage of cardiac auscultation has become ideal for automated analysis, enabling heartbeat audio recordings to serve as direct inputs for machine learning models.

While heartbeat sounds contain clinically relevant information, an open question remains as to whether such audio data can form the basis of reliable and robust predictive models, and which methodological approach provides the safest, most efficient, and most generalizable solution. It is unclear whether performance differences arise from the choice of audio representation or from the learning algorithm itself.

The objective of this study is to compare two widely used approaches for audio analysis: (i) transforming raw audio signals into visual representations and processing them with convolutional neural networks, and (ii) extracting engineered features from the audio and applying gradient boosting-based classification on them. By systematically evaluating these methods on the same datasets, this study aims to assess their relative performance, robustness, and suitability for medical assistance applications.

Possible results of this project are intended to support the development of machine learning-assisted diagnostic tools that can serve as reliable decision-support systems in cardiology, improving screening efficiency while maintaining clinical safety.

2. DATA

The dataset used in this study consists of heartbeat audio recordings collected from multiple sources and acquisition settings. The training data was collected in Northeast Brazil between 2014 and 2015 from patients aged 0–21 years during a public health campaign known as Caravana do Coração (“Caravan of the Heart”). Heart sounds were recorded using a digital phonocardiogram at various auscultation points on the chest (Aortic Valve, Mitral Valve, Pulmonary Valve, Tricuspid Valve), capturing a wide range of physiological and pathological cardiac conditions.

The separate testing data originates from the Pascal Classifying Heart Sounds Challenge (2011) and comprises two distinct subsets. The first subset includes recordings acquired using the IOS mobile application called iStethoscope, while the second consists of digital phonocardiogram (Digiscope) recordings collected in Europe. Together, these datasets provide heterogeneous recording environments and devices, enabling an evaluation of model generalization across different conditions.

2.1 Data description

The training dataset consists of heartbeat recordings collected from 1,568 patients, each contributing multiple audio observations recorded at different parts on the chest. In total, the training data contains 3163 heartbeat observations. For model development, 10% of these observations were reserved either as validation data for the convolutional neural network or as an independent evaluation set for the gradient boosting algorithm.

The target variable is binary, indicating the outcome of the heartbeat analysis as either normal (0) or abnormal (1). Class balance was explicitly addressed during data collection, resulting in an approximately equal proportion of normal and abnormal patients. This balance has been preserved when expanding patient-level data into multiple observations. The split between training and evaluation sets was performed using stratified sampling to maintain the target class distribution. Very few missing values were present in the target variable, which were dropped. Missing audio recordings were excluded during preprocessing, and all retained observations were complete and suitable for feature extraction.

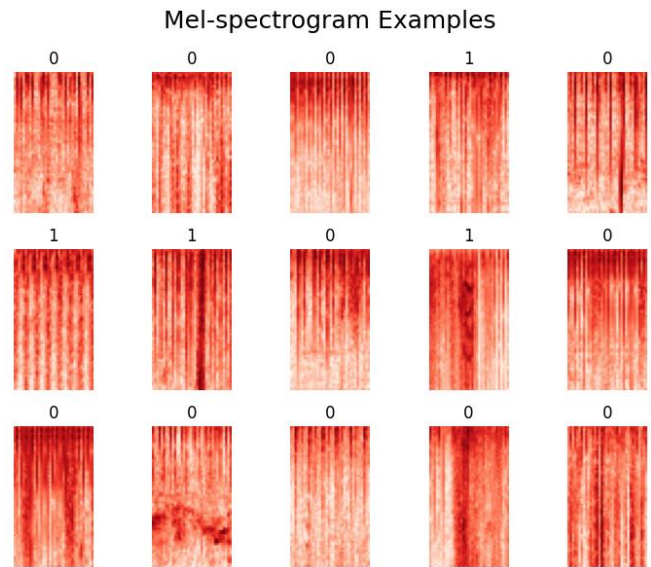
	Outcome	Filename	Location
0	1	2530_MV.wav	AV
1	1	2530_MV.wav	PV
2	1	2530_MV.wav	TV

1. Table: First Rows of the Training Data

2.2 Data understanding and preprocessing

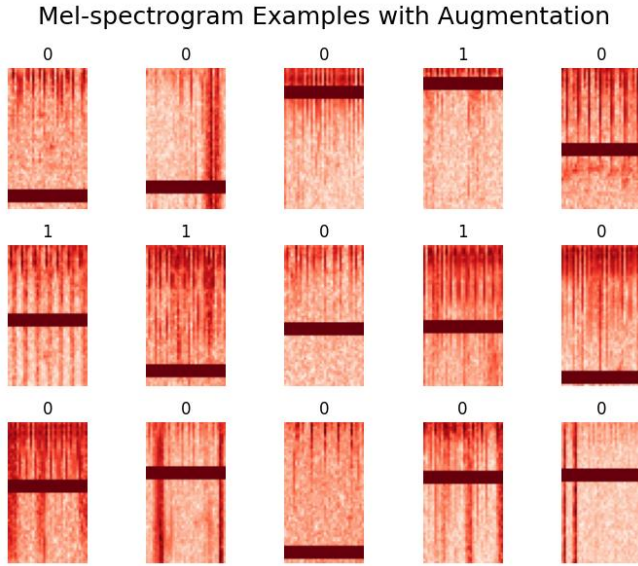
Detecting heartbeat normality from audio recordings is a nontrivial task. Recording durations vary substantially, auscultation locations differ across samples, and cardiac cycles (systole and diastole) are not temporally aligned. In addition, recordings are often affected by significant background noise and human speech. While normal heartbeats exhibit relatively consistent patterns, abnormal heartbeats may have a wide range of characteristics (like murmur, or extrasystole). Consequently, modeling performance is highly sensitive to preprocessing choices made at the initial stages of analysis.

For the convolutional neural network, raw audio recordings were loaded and transformed into Mel-spectrograms, which are time–frequency representations in which the frequency axis is non-linearly scaled to approximate human auditory perception. The transformation parameters were selected to reflect the acoustic properties of heart sounds. All recordings were cropped or zero-padded to a fixed duration of 5 seconds. The sampling rate was downsampled from 4000 Hz to 2000 Hz. A 1024-point Fast Fourier Transform (FFT) was applied, corresponding to a 512-millisecond analysis window, with a hop length of 256 samples (128 milliseconds) and a window length of 512 samples (256 milliseconds). Frequency content was restricted to the 25–600 Hz range, and 64 Mel frequency bands were computed. The resulting Mel-spectrograms were converted to the decibel scale and normalized.



1. Figure: Mel-spectrogram Examples

During evaluation and testing, audio recordings were processed using the transformations described above with center-based cropping. In contrast, during training, data augmentation was applied through a generator that performed randomly positioned 5-second cropping, added Gaussian noise to the audio signal, and randomly masked frequency bands to improve robustness.



2. Figure: Mel-spectrogram Examples with Augmentation

For the gradient boosting approach, a set of engineered audio features was extracted from each recording using Librosa library. These included time- and tempo-related features such as zero-crossing rate (measuring the frequency of sign changes in the signal), root mean square energy (representing average signal power), and the mean tempogram (capturing rhythmic content and tempo variation). Spectral features were also computed, including the spectral centroid (indicating perceived brightness) and spectral bandwidth (measuring the spread of frequency components around the centroid). In addition, mel-frequency cepstral coefficients (MFCCs) were extracted.

MFCC parameters were aligned with the Mel-spectrogram preprocessing: a sampling rate of 2000 Hz, a 1024-point FFT, a hop length of 256 samples, a window length of 512 samples, and a frequency range of 25–600 Hz. Twelve MFCC coefficients were computed, from which both the mean and standard deviation were calculated to form the final feature representation. All features were scaled based on the training set.

	zcr	rms	spec cent	tempogram
0	-0.185	0.708	-0.525	0.797
1	-0.251	1.112	-0.401	-0.710
2	1.975	9.052	1.916	-0.658

2. Table: First Rows of the Extracted Features

3. MACHINE LEARNING METHODS USED

As partly described above, this project adopted a dual approach to heartbeat classification. The primary approach employed a deep learning model based on a Convolutional Neural Network (CNN), which is well suited for processing Mel-spectrograms directly as image-like inputs. In parallel, a baseline model was established using a Gradient Boosting (GB) algorithm trained on the engineered audio features. This baseline served as a reference point for evaluating the performance and effectiveness of the CNN-based approach.

3.1 Brief description of the methods used

The final convolutional neural network (CNN) architecture consisted of a sequence of convolutional and fully connected layers designed to capture both temporal and frequency-related patterns in the input Mel-spectrograms. The first convolutional layer employed 32 filters with a kernel size of (1, 3) and a ReLU activation function to capture short-range temporal (serial) patterns. This was followed by batch normalization and a (2, 2) pooling layer. The second convolutional layer applied 64 filters with a kernel size of (3, 3) and a ReLU activation function to capture frequency-wise patterns, followed by batch normalization and a second (2, 2) pooling layer. The resulting feature maps were flattened and passed to two fully connected layers with 128 and 1 neurons, respectively, using ReLU and sigmoid activation functions. A dropout layer with a rate of 0.5 was applied between the fully connected layers to reduce overfitting. The model was implemented using TensorFlow (Keras) modules.

For the gradient boosting approach, the XGBoost algorithm was selected due to its strong performance on structured feature sets. The model was trained using the following hyperparameters: number of estimators, maximum tree depth, learning rate, subsample ratio, column subsample ratio, and the regularization parameter gamma.

3.2 Brief description of the evaluation criteria

For the convolutional neural network, model performance was evaluated using recall and the area under the receiver operating characteristic curve (AUC), while binary cross-entropy was used as the loss function.

For the gradient boosting (XGBoost) model, recall was selected as the primary optimization metric. The choice of evaluation metrics reflects the clinical motivation of the task: minimizing false negatives is critical, as failing to detect abnormal heartbeats poses a greater risk than incorrectly flagging normal cases. Accordingly, recall was prioritized to emphasize this sensitivity.

4. EXPERIMENTS

This section describes the model tuning procedures, evaluation, and a comparative analysis of the results of the convolutional neural network and gradient boosting model.

4.1 Tuning

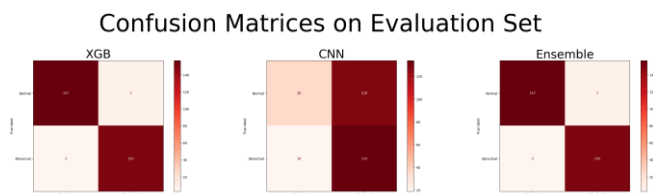
For the convolutional neural network, optimization was performed using the Adam optimizer, with the learning rate adapted automatically during training. Early stopping was applied to prevent overfitting, monitoring the recall metric on the validation set with a patience of 3 epochs. Training was conducted using the augmented data generator described in Section 2.2, while the held-out evaluation set was used exclusively for validation.

For the gradient boosting approach, hyperparameters were optimized using the Optuna framework. A total of 30 optimization trials were conducted, each evaluated using 5-fold cross-validation with a binary logistic objective. The tuning process explored parameters including the number of estimators, maximum tree depth, learning rate, subsampling ratio, column subsampling ratio, and the regularization parameter gamma. Recall was used as the optimization metric to align with the clinical objective of minimizing false negatives.

4.2 Evaluation

Model performance was evaluated on a held-out evaluation set and on two independent test datasets acquired under different recording conditions. In addition to the individual convolutional neural network and gradient boosting models, an ensemble model was evaluated. The ensemble combined the two approaches by computing the mean of their predicted class probabilities, with final class predictions obtained using a decision threshold of 0.5.

On the evaluation set, the results differed markedly across modeling approaches. The CNN achieved low performance overall (labeling most samples as abnormal), with high recall for abnormal samples (0.92) but very low recall for normal samples (0.15), resulting in an overall accuracy of 52%. In contrast, the XGBoost model performed excellently, achieving balanced precision and recall for both classes (0.97–0.98) and an overall accuracy of 97%. The ensemble model mirrored XGBoost’s strong performance, maintaining 97% accuracy and balanced F1-scores, indicating that while the models capture partially complementary information, CNN contributed little to no additional value in this case.

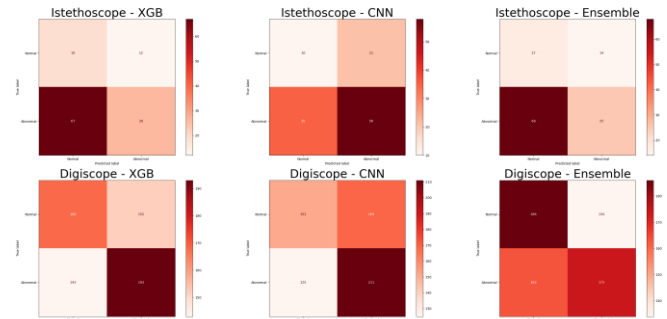


3. Figure: Confusion Matrices on Evaluation Set

Evaluation on the mobile application data (iStethoscope) test set revealed notable differences in generalization. The CNN achieved moderate overall performance with an accuracy of 51%, showing moderate recall for abnormal samples (0.56) but low recall for normal samples (0.35), whereas the XGBoost model achieved lower overall accuracy (41%), performing slightly better on normal recall (0.71) but worse on abnormal recall (0.31). The ensemble model yielded an intermediate accuracy of 34%, indicating that averaging probability outputs can mitigate some weaknesses specific to individual models, though overall performance remained very limited on this dataset.

On the Digiscope test set, which consists of digital phonocardiogram recordings acquired in a clinical setting, the CNN and XGBoost models both achieved similarly low overall accuracy (53-55%), with the CNN slightly favoring abnormal recall (0.70). The ensemble model resulted in comparable performance, suggesting that combining predictions provides limited additional benefits, and th had very low performance in this case, but both models demonstrate robustness across recording devices.

Confusion Matrices on iStethoscope and DigiScope



4. Figure: Confusion Matrices on Test Sets

Across the evaluation sets, the ensemble model generally mirrored the performance of the stronger individual classifier, providing more stable predictions than the CNN, which struggled on normal beats. On the iStethoscope set, XGBoost performed very poorly overall, but captured normal beats slightly better, whereas the CNN favored abnormal beats, leading to moderately better overall performance. The ensemble moderated these extremes but did not substantially improve overall accuracy in this noisy, mobile-device setting. On the Digiscope set, both CNN and XGBoost achieved similar accuracy (55%) with small differences in class-wise recall and precision, and the ensemble produced comparable performance. These results suggest that while CNN and XGBoost capture partially complementary information, the benefit of ensembling may be dataset-dependent: it can stabilize predictions but adds little advantage when the individual models are already balanced, as in high-quality clinical recordings.

5. CONCLUSION

This study compared convolutional neural network trained on Mel-spectrograms and gradient boosting models trained on engineered audio features for heartbeat classification across heterogeneous datasets. On the evaluation set, XGBoost achieved the highest performance (97% accuracy, balanced precision and recall), while the CNN favored Abnormal beats (recall 0.92) but struggled with normal beats (recall 0.15), resulting in moderate overall accuracy (52%). The ensemble closely mirrored XGBoost, indicating partial complementarity between the models but limited added value from the CNN. On iStethoscope mobile recordings, the CNN maintained moderately low performance (51% accuracy) with higher abnormal recall but low normal recall, whereas XGBoost performed poorly (41%), and the ensemble slightly improved class balance. On Digiscope clinical recordings, both models achieved similar accuracy (53-55%), with the ensemble offering minimal additional benefit, suggesting robustness across high-quality recording devices.

These results highlight that feature-based models can outperform end-to-end CNNs in noisy or limited-data settings, while CNNs may be sensitive to device-specific artifacts. Ensembling stabilizes predictions but is most useful when individual models provide complementary strengths. The poor performance of the models is largely attributable to suboptimal parameter choices, which require further experimentation. Additionally, the selection of the Mel-spectrogram representation of heartbeats may not fully capture the relevant cardiac signal characteristics, limiting the models' ability to discriminate subtle abnormalities. For XGBoost, performance could be significantly enhanced by incorporating more insightful, domain-informed features, possibly hand-crafted features developed collaboratively by cardiologists and data scientists: allowing the models to leverage both physiological knowledge and data-driven patterns for improved predictive accuracy. Further future work could improve CNN robustness through hybrid feature approaches, supporting the development of reliable, device-agnostic machine learning tools for cardiac auscultation represented by audio.

6. CLASSIFICATION REPORTS

	precision	recall	f1	support
XGB				
Normal	0.98	0.98	0.98	164
Abnormal	0.97	0.97	0.97	153
accuracy			0.97	317
macro avg.	0.97	0.97	0.97	317
weighted avg.	0.97	0.97	0.97	317
CNN				
Normal	0.67	0.15	0.24	164
Abnormal	0.50	0.92	0.65	153
accuracy			0.52	317
macro avg.	0.58	0.53	0.44	317
weighted avg.	0.59	0.52	0.44	317

3. Table: Classification Reports on Evaluation Set

	precision	recall	f1	support
XGB				
Normal	0.26	0.71	0.38	31
Abnormal	0.76	0.31	0.44	93
accuracy			0.41	124
macro avg.	0.51	0.51	0.41	124
weighted avg.	0.64	0.41	0.43	124
CNN				
Normal	0.21	0.35	0.27	31
Abnormal	0.72	0.56	0.63	93
accuracy			0.51	124
macro avg.	0.47	0.46	0.45	124
weighted avg.	0.59	0.51	0.54	124

4. Table: Classification Reports on Mobile Data

	precision	recall	f1	support
XGB				
Normal	0.52	0.53	0.52	320
Abnormal	0.54	0.54	0.54	336
accuracy			0.53	656
macro avg.	0.53	0.53	0.53	656
weighted avg.	0.53	0.53	0.53	656
CNN				
Normal	0.57	0.42	0.53	320
Abnormal	0.56	0.70	0.56	336
accuracy			0.55	656
macro avg.	0.55	0.55	0.55	656
weighted avg.	0.55	0.55	0.55	656

5. Table: Classification Reports on Clinical Data

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